SHARAHBILL MOHAMED ABDI Data & Machine Learning Engineer | MEAL & Research

I build forecasting and machine learning tools, and run the research that shows *what is working* in development programmes.

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95.1%	0.889	8	Top 10%
explained variance on drought forecasts	ROC AUC on flood risk for 1,450 Kenyan wards	MEAL and research assignments in 2025 and 2026	Kaggle Data Visualization Challenge

PROFILE

I am a data and machine learning engineer, and I also lead MEAL and research at Cortech Consulting Group. On the research side I design the frameworks and KoboToolbox surveys for client evaluations, analyse the data in Python and write the reports. On the engineering side I build forecasting, classification and computer vision models and put them online as FastAPI and Streamlit apps.

EXPERIENCE

MEAL Officer / Research

Cortech Consulting Group | Nairobi, Kenya — 2025 to Present

- Lead the MEAL and research work on the firm's 2025 and 2026 projects, including the **8 assignments** below.
- Set up MEAL frameworks so every assignment has a clear plan for monitoring, evaluation and learning.
- Run household and community surveys in KoboToolbox, then clean, analyse and model the data in Python.
- Write the evaluation and research reports that go to government, multilateral and NGO clients.

SELECTED ASSIGNMENTS

- 2026 **Federal Government of Somalia** (Ministry of Agriculture & Irrigation): early warning and DRR training, with system design
- 2026 **IGAD**: end-of-project evaluation of "Improving Land Governance in the IGAD Region" (Phase II)
- 2025 **World Bank**: capacity assessment of Somalia's Federal Member State Ministries of Health
- 2025 **IOM & African Union**: monitoring and implementation framework for the Free Movement of Persons Protocol
- 2025 **Heifer International Kenya**: agritech landscape study on scaling digital agriculture
- 2025 **Danish Refugee Council**: mapping financial service providers for refugees and host communities
- 2025 **Qatar Red Crescent**: SME financing capacity building in Somalia
- 2025 **Welthungerhilfe**: evaluation of "Chukua Control"

Data Analysis Trainer (Part-Time)

Trainify Institute of Technology | Nairobi, Kenya (Remote / Hybrid)

- Taught Python data analysis (Pandas, NumPy, visualization) to **multiple cohorts** of students.
- Mentored them through full projects, from raw data to a final presentation.

EDUCATION

BSc in Mathematics and Computing
The Open University of Kenya

Data Science
Moringa School, April 2026

Data Visualization with Python
Moringa School, 2025

ACHIEVEMENTS & CERTIFICATIONS

Top 10%, Kaggle Data Visualization Challenge

Member of the best team, Nairobi Flood Guard AI, named best project of 2026

Best Problem Solver, Moringa School Bootcamp

ICDL Certification

LANGUAGES

Somali (Native)

English (Fluent)

Kiswahili (Fluent)

PROJECTS

Nairobi Flood Guard AI | Geospatial ML, Team Project

Best project of 2026

`github.com/t31sha/Nairobi_Flood_Guard-AI` | `Live: nairobi-flood-guard.streamlit.app`

- Flood early warning and matatu rerouting for **1,450 Kenyan wards**, with SMS alerts.
- Our calibrated XGBoost hit **ROC AUC 0.889** and **recall 0.80** on county-held-out spatial cross-validation.
- It beat logistic regression, random forest and neural network models we tested against it.

Tools: Python, GeoPandas, OSMnx, XGBoost, FastAPI, Streamlit, Folium, Docker, GitHub Actions

SPEI Drought Forecasting Platform | Time Series Forecasting

`github.com/t31sha/Drought-Forcasting_Prediction` | `Live: drought-intelligence.streamlit.app`

- Forecasts SPEI-12 drought 1, 3 and 6 months ahead with **95.1% explained variance** and **98.3% directional accuracy**.
- Optuna-tuned XGBoost trained on **76 years** of monthly climate data from Oran, Algeria, with SHAP and bootstrap intervals.

Tools: Python, pandas, scikit-learn, XGBoost, Optuna, SHAP, Streamlit, Plotly

CassavaWatch | Computer Vision, Full-Stack ML

`github.com/t31sha/CassavaWatch` | `Live: cassavawatch.onrender.com`

- Fine-tuned MobileNetV3Small to **75.2% accuracy**, then shrank it to a **1.11 MB** TFLite model that fits a 512 MB free-tier server.
- Diagnoses cassava disease from 3 to 6 leaf photos, with English or Swahili advice, offline queueing and outbreak alerts.

Tools: Python, TensorFlow / Keras, TFLite, FastAPI, SQLite, Leaflet, Qwen API, pytest, Render

Customer Churn Prediction | Machine Learning

`github.com/t31sha/Customer-Churn_Prediction`

- Random Forest tuned for recall on **3,333 telecom records** (about 15% churn), with a risk export ready for Tableau.
- Biggest churn drivers: daytime charges, 3+ service calls and international plans.

Tools: Python, pandas, scikit-learn, Tableau

NSE Stock Price Forecasting | Time Series Forecasting

`github.com/t31sha/NSE--Stock-Market_-Predictions`

- Predicted Safaricom (SCOM) prices from 2025 NSE data with an **MAE of KES 0.60**, using lag and calendar features.
- Kept a chronological 80/20 split so the Random Forest never saw future prices.

Tools: Python, pandas, scikit-learn, Matplotlib, Seaborn

Mobile User Behavior Classification | Machine Learning

`github.com/t31sha/User-Behaviour_Prediction`

- Sorted **700 mobile users** into five behavioral tiers from device telemetry with a Random Forest.
- App usage time (25.6%), battery drain (22.8%) and app count (20.3%) mattered most.

Tools: Python, pandas, NumPy, scikit-learn, Seaborn, Matplotlib

SKILLS

Machine Learning	Classification, Regression, Time Series Forecasting, Feature Engineering, Model Tuning, CRISP-DM; scikit-learn, XGBoost, Optuna, SHAP
Deep Learning	Computer Vision, Transfer Learning; TensorFlow / Keras, TFLite (model quantization)
Deployment & APIs	FastAPI, REST APIs, Streamlit, Render, Docker, GitHub Actions, pytest, LLM API integration (Qwen)
Programming & Data	Python, Pandas, NumPy, SQLite, SQLAlchemy, Git, GitHub, Jupyter Notebook
Analysis & Viz	Statistical Analysis, EDA; Tableau, Plotly, Matplotlib, Seaborn; GeoPandas, OSMnx, Folium
MEAL & Research	MEAL Frameworks, Survey Design, Household & Community Surveys, KoboToolbox, Evaluation Reporting